A Note on Universality of 1-Degenerate n-Vertex r-Graphs

Sifei Li

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1 Summary

A graph Γ is $\mathcal{H}$-universal if it contains a copy of every $H \in \mathcal{H}$. This note studies r-uniform hypergraphs that are universal for the class $\mathcal{H}^{(r)}(n,1)$ of n-vertex 1-degenerate r-graphs. It extends the problem considered in [ABK25], where the authors studied universality for n-vertex D-degenerate r-graphs with $D \geq 2$, to the case $D = 1$.

1.1 Recall from the original paper

The original paper proves upper bounds for $\mathcal{H}^{(r)}(n,D)$-universal r-graphs in the case $D > 1$. A key part of the construction is the random r-graph $\Gamma(r,n,D)$. In particular, Proposition 4.2(c) of [ABK25] states that $\Gamma(r,n,D)$ has at most

$$200 \cdot 3^D rn$$

vertices and asymptotically almost surely has at most

$$2(20 \cdot 3^D r)^r (\log\log n)^{2r+1} (\log n)^{2/D} n^{r-1/D}$$

edges.

In this note, we treat $D = 1$ as an analogous case. The main point is to modify the random block model from the original paper so that the embedding argument can still be carried out.

1.2 New result

The main result of this note is the following $D = 1$ analogue. In the random model used in this note, the corresponding version of Proposition 4.2(c) is as follows: the random r-graph $\Gamma = \Gamma(r,n,1)$ has at most

$$150 \cdot 3^{r-1} rn$$

vertices and asymptotically almost surely has at most

$$2(100 \cdot 3^{r-1} r)^r (\log\log n)^r (\log n)^2 n^{r-1}$$

edges. Notice that if we plug directly in $D = 1$ into the bound from the original paper, it is $O((\log\log n)^{2r+1}(\log n)^2 n^{r-1})$. The power of $\log\log n$ is larger, so we actually achieved a slightly smaller bound.

1.3 Key ideas of the proof

In the original proof, the parameter D plays two different roles. First, it is the combinatorial degeneracy parameter in the class $\mathcal{H}^{(r)}(n,D)$. Second, it controls the scale of the random block model and therefore the final edge bound. When $D = 1$, the first role does not seem to cause serious difficulty, but the second one does: substituting directly $D = 1$ into the original block model no longer gives the required separation between consecutive block scales. The main idea is therefore to separate these two roles. We keep $D = 1$ as the degeneracy parameter, but introduce a new parameter

$$q := r - 1$$

to control the block scales. Throughout the note, we assume $r \geq 3$, so that $q \geq 2$. The modified model uses block parameters satisfying the following:

$$\Delta_{k-1} = \Delta_k^q.$$

The rest of the proof follows the structure of the original paper. We consider only $r \geq 3$. When $r = 2$, 1-degenerate graphs are forests. Universal graphs with $O(n \log n)$ edges are known. Chung and Graham claimed such a construction in 1983 [CG83]; an error in their proof was subsequently identified, and corrected and improved constructions are now available [KK25].

1.4 Further directions

The bound obtained here may not be optimal: in the case $r = 2$ it does not recover the best known bound for graphs [KK25], which indicates this result may admit further improvement.

2 Bounds for $\mathcal{H}^{(r)}(n, 1)$-universal r-graph

In this section, we provide the full definitions and proofs for the upper bound of edges and vertices of the hypergraph Γ, which is universal for the class of 1-degenerate n-vertex r-graphs.

2.1 Preliminaries

All unspecified logarithms are base 2. Whenever $\log n$ is used as an integer index or as the number of sub-blocks, it denotes $\lceil \log_2 n \rceil$. We suppress the floor and ceiling signs needed to make all block and sub-block cardinalities integral. These harmless roundings affect only the constant factors in the estimates below.

For a set S, let $\binom{S}{r}$ denote the collection of subsets with r elements in S. For an r-graph H, let $E(H)$ denote the collection of hyperedges of H, and $V(H)$ to be the collection of vertices of H. For a vertex $v \in V(H)$, define its link $L(v) = \{S \in \binom{V(H)}{r-1} | v \cup S \in E(H)\}$. If $V(H)$ has an order $v_1, ..., v_n$ for some integer n, then we define the back-link of a vertex v_i to be $L^-(v_i) = L(v_i) \cap \binom{\{v_1,...,v_{i-1}\}}{r-1}$.

A r-graph H is said to be D-degenerate if there exists an ordering $v_1, ..., v_n$ of $V(H)$ such that $|L^-(v_i)| \leq D$ for every i.

Theorem 1 (Chernoff bound)**.** *[MU17] Let $X = \sum_{i=1}^n X_i$, where the X_i are independent Bernoulli random variables. Then*

1. $\mathbb{P}(X \geq (1+\delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2+\delta}\mathbb{E}X\right)$ *for all* $\delta > 0$,
2. $\mathbb{P}(X \leq (1-\delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2}\mathbb{E}X\right)$ *for all* $0 < \delta < 1$.

Observation 2.1 in the paper [ABK25] still holds for $D = 1$:

Observation 1. *Let H be a 1-degenerate r-graph, and let X be the set of vertices in H of degree at least k, with $k > 0$. Then $|X| \leq rn/k$.*

Proof. Let H be an n-vertex 1-degenerate r-graph, let $v_1, \ldots, v_n$ be a 1-degeneracy ordering. We count the edges in H from v_n. By definition, v_n is contained in at most 1 edge. For any $1 < m \leq n$, after counting the edges that include $v_m, \ldots, v_n$, v_{m-1} is contained in at most one more edge that has not been counted. Thus H has at most n edges. We have $|E(H)| \leq n$.

Now let

$$X := \{v \in V(H) : d_H(v) \geq k\},$$

where $d_H(v)$ is the degree of vertex v in H. Since every vertex in X has degree at least k,

$$k|X| \leq \sum_{x \in X} d_H(x) \leq \sum_{v \in V(H)} d_H(v) = r|E(H)| \leq rn,$$

rearranging gives

$$|X| \leq \frac{rn}{k}.$$

□

2.2 Proofs

Definition 1 (The block model for the case $D = 1$)**.** *Fix $r \geq 3$, and let $q := r - 1$. Let N be the smallest integer such that*

$$n^{q^{1-N}} \leq 3^{q^2}.$$

For $1 \leq k \leq N$, define

$$\Delta_k := n^{q^{-k}},$$

and set $\Delta_0 := n$, so $\Delta_{k-1} = \Delta_k^q$. Let $W = W_1 \sqcup \cdots \sqcup W_N$ be a disjoint union of blocks satisfying

$$|W_k| = \frac{100 \cdot 3^q rn}{\Delta_k} \qquad \text{for each } 1 \leq k \leq N.$$

Each block W_k is further partitioned into sub-blocks

$$W_k = W_{k,1} \sqcup \cdots \sqcup W_{k,\log n},$$

where

$$|W_{k,1}| = \frac{1}{2}|W_k| \qquad \text{and} \qquad |W_{k,j}| \geq \frac{|W_k|}{2 \log n} \quad \text{for all } j \geq 2.$$

For $s \in \binom{W}{r}$, let $\pi_i(s) := |s \cap W_i|$, and define

$$p^* := K_r \frac{(\log n)^2}{n}, \qquad p_s := \min\left\{1,\, p^* \prod_{i=1}^{N} \Delta_i^{\pi_i(s)}\right\},$$

with a fixed constant K_r satisfying

$$K_r \geq \frac{16 \ln 2}{25 \cdot 3^{r-1}} \left(1 + \frac{5}{r}\right).$$

For definiteness, we may take $K_r = 1$. The random r-graph Γ on vertex set W is obtained by including each $s \in \binom{W}{r}$ as its edge independently with probability p_s.

Proposition 1. *With the notation of Definition 1, the following holds for every fixed $r \geq 3$ and all sufficiently large n*

- $N < \log \log n$

Proof. From the definition of N in Definition 1, we have $n^{q^{1-(N-1)}} > 3^{q^2}$. Taking $\log_3$ and $\log_q$ on both sides gives $(2 - N) + \log_q \log_3 n > 2$. Rearranging the inequality gives:

$$N < \log_q \log_3 n \leq \log \log n$$

as desired. □

Proposition 2 (The $D = 1$ version of Proposition 4.2(c))**.** *For every fixed $r \geq 3$, there is a constant $C_r > 0$ such that, for all sufficiently large n, the random r-graph $\Gamma = \Gamma(r, n, 1)$ defined in Definition 1 has at most $150 \cdot 3^q rn = 150 \cdot 3^{r-1} rn$ vertices, and asymptotically almost surely has at most*

$$C_r (\log \log n)^r (\log n)^2 n^{r-1}$$

edges.

Proof. We first bound the number of vertices. By the definition of N, we have

$$\Delta_{N-1} = n^{q^{1-N}} \leq 3^{q^2} \text{ and } \Delta_{N-2} = n^{q^{2-N}} > 3^{q^2}.$$

Raise the first inequality to the power $1/q$ and combine with the second inequality,

$$3^q \geq \Delta_{N-2}^{1/q^2} = \Delta_N > 3.$$

Moreover, using the property $\Delta_{k-1} = \Delta_k^q$ for $1 \le k \le N$, and $q = r-1 \ge 2$, we get

$$|W| = |W_1| + |W_2| + \cdots + |W_N| = 100 \cdot 3^q rn \sum_{k=1}^{N} \frac{1}{\Delta_k} = 100 \cdot 3^q rn \sum_{s=0}^{N-1} \frac{1}{\Delta_N^{q^s}}$$
$$\le 100 \cdot 3^q rn \sum_{s=0}^{\infty} \frac{1}{\Delta_N^{q^s}} \le 100 \cdot 3^q rn \sum_{s=0}^{\infty} \frac{1}{3^{2^s}} = 100 \cdot 3^q rn \frac{1}{1-\frac{1}{9}} < 150 \cdot 3^q rn.$$

Thus Γ has at most $150 \cdot 3^q rn$ vertices.

We now estimate the expected number of edges. Fix an ordered r-tuple

$$S = (i_1, \ldots, i_r) \in [N]^r.$$

For each $k \in [N]$, let $m_S(k)$ be the multiplicity of k in S. The expected number of edges of Γ with one vertex in each of $W_{i_1}, \ldots, W_{i_r}$, denoted by $\mathbb{E}\big[|E(W_{i_1}, \ldots, W_{i_r})|\big]$,is at most

$$\min\left\{1, p^* \prod_{\ell=1}^{r} \Delta_{i_\ell}\right\} \prod_{k \in [N]} |W_k|^{m_S(k)} \le p^* \prod_{\ell=1}^{r} \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)}.$$

Thus

$$\mathbb{E}\big[|E(W_{i_1}, \ldots, W_{i_r})|\big] \le p^* \prod_{\ell=1}^{r} \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)} = p^* \prod_{\ell=1}^{r} \Delta_{i_\ell} \frac{(100 \cdot 3^q rn)}{\Delta_{i_l}} = p^* \prod_{\ell=1}^{r} 100 \cdot 3^q rn.$$

Summing over all N^r choices of r-tuples and using $N \le \log\log n$ from Proposition 1, we get

$$\mathbb{E}|E(\Gamma)| \le N^r p^* \prod_{\ell=1}^{r} 100 \cdot 3^q rn = N^r p^* (100 \cdot 3^q rn)^r \le (\log\log n)^r \frac{K_r (\log n)^2}{n} (100 \cdot 3^q rn)^r := C_r' (\log\log n)^r (\log n)^2 n^{r-1},$$

where $C_r' = K_r (100 \cdot 3^q \cdot r)^r$.

It remains to turn this expectation bound into an asymptotically almost sure bound. Denote $M = C_r' (\log\log n)^r (\log n)^2 n^{r-1}$ and apply part 1 of Theorem 1 (Chernoff bound) with

$$\delta = 2\frac{M}{\mathbb{E}|E(\Gamma)|} - 1,$$

notice that $\delta \ge 1$, we get

$$\mathbb{P}[|E(\Gamma)| \ge 2M] = \mathbb{P}(|E(\Gamma)| \ge (1+\delta)\mathbb{E}|E(\Gamma)|)$$
$$\le exp[-\frac{\delta^2}{2+\delta}\mathbb{E}|E(\Gamma)|] \le exp[-\frac{\delta \mathbb{E}|E(\Gamma)|}{3}] \le exp[-M/3] = o(1)$$

Thus we may take

$$C_r := 2K_r (100 \cdot 3^q r)^r = 2(100 \cdot 3^{r-1} r)^r,$$

where in the final equality we used $K_r = 1$ and $q = r-1$. This completes the proof. □

Definition 2 (Embedding Strategy)**.** *[ABK25] Given a 1-degenerate r-graph H, let $v_1, \ldots, v_n$ be a 1-degeneracy ordering of $V(H)$, and let Γ have the properties of Proposition 2. Our embedding strategy is defined by the iterative construction of a partial embedding $\psi_t : \{v_1, \ldots, v_t\} \to \Gamma$. We start with ψ_0, the trivial partial embedding of no vertices into Γ.*

For each $1 \le t \le n$, we say a vertex $u \in V(\Gamma)$ is a candidate for v_t if $\psi_{t-1}(L^-(v_t)) \subseteq L(u)$. We say such a vertex $u \in \Gamma$ is an available candidate if in addition $u \notin \mathrm{Im}(\psi_{t-1})$. Let k be minimal such that $\Delta_k < \deg(v_t)$. If there is no such k choose $k = N$. Choose j minimal such that there is some $u \in W_{k,j}$ which is an available candidate for v_t. We define $\psi_t = \psi_{t-1} \cup \{v_t \to u\}$. If there is no available candidate for v_t in W_k, we say ψ_t and the subsequent partial embeddings do not exist and that the embedding strategy fails at time t. If the embedding strategy does not fail, then $\psi := \psi_n$ gives an embedding of H into Γ.

For $D = 1$, we use the same embedding strategy and the corresponding specialization of the well-behaved multiset definition.

Definition 3 (Well-behaved multisets for the case $D = 1$)**.** *Let $\mathcal{B} = (B_1, \dots, B_t)$ be a multiset such that each $B_i \subseteq \binom{W}{r-1}$. For B_i, write*

$$V(B_i) := \bigcup_{b \in B_i} b,$$

and write

$$V(\mathcal{B}) := \bigcup_{i=1}^{t} V(B_i).$$

We say that $\mathcal{B}$ is well-behaved if the following three conditions hold:

$$|B_i| \le 1 \qquad \text{for all } i \in [t], \tag{WB1}$$

$$\big|\{\, i \in [t] : u \in V(B_i) \,\}\big| \le \Delta_{k-1} \qquad \text{for all } 1 \le k \le N \text{ and all } u \in W_k, \tag{WB2}$$

and

$$|V(\mathcal{B}) \cap W_{k,j}| \le \frac{1}{2}|W_{k,j}| \qquad \text{for all } 1 \le k \le N \text{ and all } 1 \le j \le \log n. \tag{WB3}$$

Lemma 1. *Let $r \ge 3$, let n be sufficiently large, and let $t \in [n]$. Let Γ be the random block model defined in Definition 1, and let $\mathcal{B} = (B_1, \dots, B_t)$ be a well-behaved multiset of size t. Fix $1 \le k \le N$ and $v \in W_k \setminus V(\mathcal{B})$. Let $\mathcal{E}_v$ be the event that there exists some $B_i \in \mathcal{B}$ such that $B_i \subseteq L(v)$. Then*

$$\mathbb{P}(\mathcal{E}_v) \ge \min\left\{\frac{1}{2},\, \frac{1}{2}\, tp^*\Delta_k\right\}.$$

Proof. Let $1 \le k \le N$, and $v \in W_k \setminus V(\mathcal{B})$. If $B_i = \emptyset$ for some $i \in [t]$, then $B_i \subseteq L(v)$ always holds, and hence $\mathbb{P}(\mathcal{E}_v) = 1$. We may therefore assume that $B_i \ne \emptyset$ for every $i \in [t]$. By (WB1), for each $i \in [t]$, there is an $(r-1)$-set $b_i \in \binom{W}{r-1}$ such that $B_i = \{b_i\}$.

Let $S := \{b_i : i \in [t]\}$ be the set of distinct $(r-1)$-sets appearing among the b_i's, and for $b \in S$, let $m_b := |\{\, i \in [t] : b_i = b \,\}|$ be the multiplicity of b, and we have $\sum_{b \in S} m_b = t$.

Fix $b \in S$, and write $b = \{x_1, \dots, x_{r-1}\}$, where $x_l \in W_{k_l}$ for $1 \le l \le r-1$. Condition (WB2) gives

$$m_b \le |\{i \in [t] : x_l \in V(B_i)\}| \le \Delta_{k_\ell - 1} \qquad \text{for each } 1 \le l \le r-1.$$

For each $1 \le k_l \le N$, $\Delta_{k_l} = \Delta_{k_l-1}^{1/q} \ge m_b^{1/q}$. Therefore

$$\prod_{l=1}^{r-1} \Delta_{k_l} \ge m_b^{(r-1)/q} = m_b.$$

Let E_b be the event that $b \in L(v)$. Since $v \notin V(\mathcal{B})$, the set $\{v\} \cup b$ has size r, and E_b is exactly the event that $\{v\} \cup b$ is an edge of Γ. Therefore

$$\mathbb{P}(E_b) = p_{\{v\} \cup b} = \min\left\{1,\, p^*\Delta_k \prod_{\ell=1}^{q} \Delta_{k_\ell}\right\} \ge \min\{1,\, p^*\Delta_k m_b\}.$$

The events $(E_b)_{b \in S}$ are independent, since they concern distinct r-sets of the form $\{v\} \cup b$. Thus, $\mathcal{E}_v = \bigcup_{b \in S} E_b$, and $\mathbb{P}(\mathcal{E}_v) = 1 - \prod_{b \in S}\big(1 - \mathbb{P}(E_b)\big)$.

Furthermore, notice that

$$\sum_{b \in S} \mathbb{P}(E_b) \ge \sum_{b \in S} \min\{1,\, p^*\Delta_k m_b\} \ge \min\left\{1,\, p^*\Delta_k \sum_{b \in S} m_b\right\} = \min\{1,\, tp^*\Delta_k\}.$$

Since $1-x \le e^{-x}$ for all $x \in [0,1]$ (which follows from the Taylor expansion $e^{-x} = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} = 1 - x + \frac{x^2}{2} - \cdots \ge 1 - x$), we have

$$\mathbb{P}(\mathcal{E}_v) \ge 1 - \prod_{b \in S} \exp\left(-\mathbb{P}(E_b)\right) = 1 - \exp\left(-\sum_{b \in S} \mathbb{P}(E_b)\right) \ge 1 - \exp\bigl(-\min\{1,\, tp^*\Delta_k\}\bigr).$$

Finally, notice that for every $x \ge 0$, $1 - e^{-\min\{1,x\}} \ge \frac{1}{2}\min\{1, x\}$, which can be verified by considering two cases: If $x \le 1$, then $1 - e^{-x} \ge x - \frac{x^2}{2} \ge \frac{x}{2}$ by Taylor expansion again; if $x \ge 1$, then $1 - e^{-1} > \frac{1}{2}$. Applying this with $x = tp^*\Delta_k$, we get

$$\mathbb{P}(\mathcal{E}_v) \ge \frac{1}{2}\min\{1,\, tp^*\Delta_k\} = \min\left\{\frac{1}{2},\, \frac{1}{2}\, tp^*\Delta_k\right\},$$

as required. □

Lemma 2. *Let $r \ge 3$ and n be sufficiently large. Then the random r-graph Γ defined in the Definition 1 a.a.s. has the following property: $\forall t \le n$ and well-behaved multiset $\mathcal{B}$ of size t. $\forall k \le N, j \le \log n$, we have*

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\}| \ge \min\left\{\frac{1}{16}, \frac{tp^*\Delta_k}{8}\right\} |W_{k,j}| \tag{1}$$

Proof. Let r, n, k, j be integers. $r \ge 3$, n large enough. Let $t \le n$ and $\mathcal{B}$ be a well-behaved multiset of size t. Fix $1 \le k \le N$ and $1 \le j \le \log n$.

It suffices to only prove the lemma under the assumption $tp^*\Delta_k \le 1$. If $tp^*\Delta_k > 1$, we can pick a t' such that $\frac{1}{2} < t'p^*\Delta_k \le 1$, and arbitrarily choose $\mathcal{B}' \subseteq \mathcal{B}$ with $|\mathcal{B}'| = t'$. Notice that if $\mathcal{B}$ is a well-behaved multiset, then its submultiset $\mathcal{B}'$ is also a well-behaved multiset. Then if the lemma holds under the assumption $tp^*\Delta_k \le 1$, we have

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}' \subseteq \mathcal{B}, B_i \subseteq L(u)\}| \ge \frac{1}{16}|W_{k,j}|$$

as desired. Notice that such t' exists as $p^*\Delta_k$ is small enough, indeed, $p^*\Delta_k \le K_r(\log n)^2 n^{(1/q)-1} = o(1)$ with $q \ge 2$.

Now assume $tp^*\Delta_k \le 1$. If there exists an i such that $B_i = \emptyset$, then $B_i \subseteq L(u)$ for every $u \in W_{k,j}$, and the desired inequality is immediate.

Let $U := W_{k,j} \setminus V(\mathcal{B})$. By (WB3), $U \ge \frac{1}{2}|W_{k,j}|$. For each $u \in U$, let X_u be the indicator of the event $\mathcal{E}_u$ that there exists $i \in [t]$ such that $B_i \subseteq L(u)$, and set $X := \sum_{u \in U} X_u$.

The random variables $(X_u)_{u \in U}$ are independent. Indeed, X_u depends only on edge indicators of sets of the form $\{u\} \cup b$, where b is the unique element of some B_i. Since each $u \in W_{k,j} \setminus V(\mathcal{B})$ does not lie inside any link set $b \in B_i \in \mathcal{B}$, the r-sets $\{u\} \cup b$ used for different u's are distinct. By definition, the edge indicator is an independent random variable, therefore the family $(X_u)_{u \in U}$ is mutually independent.

By Lemma 1, for every $u \in U$, $\mathbb{P}(\mathcal{E}_u) \ge \min\{\frac{1}{2}, \frac{1}{2}tp^*\Delta_k\}$. Since we assumed $tp^*\Delta_k \le 1$ at the beginning, this gives $\mathbb{P}(\mathcal{E}_u) \ge tp^*\Delta_k/2$. Therefore,

$$\mathbb{E}X = \sum_{u \in U} \mathbb{E}X_u \ge \frac{1}{2}tp^*\Delta_k|U| \ge \frac{1}{4}tp^*\Delta_k|W_{k,j}| := 2T. \tag{2}$$

where $T = \frac{1}{8}tp^*\Delta_k|W_{k,j}|$.

Now we show that the probability that the conclusion fails is $o(1)$. By part 2 of Theorem 1,

$$\mathbb{P}(X \le T) \le \mathbb{P}\left(X \le \left(1 - \frac{1}{2}\right)\mathbb{E}X\right) \le \exp\left(-\frac{1}{8}\mathbb{E}X\right).$$

To obtain a lower bound on $\mathbb{E}X$, note that

$$|W_{k,j}| \ge \frac{|W_k|}{2\log n} = \frac{50 \cdot 3^q rn}{\Delta_k \log n},$$

therefore

$$\mathbb{E}X \geq \frac{1}{4}tp^*\Delta_k|W_{k,j}| \geq \frac{1}{4}tp^*\Delta_k\frac{50\cdot 3^q rn}{\Delta_k \log n} \geq 25/2\cdot 3^q r K_r\, t\log n.$$

Combining all together and using $\log n = \frac{\ln n}{\ln 2}$,

$$\mathbb{P}(X<T) \leq \exp\left(-\frac{1}{8}\cdot 25/2\cdot 3^q r K_r\, t\log n\right) = n^{-25\cdot 3^q\cdot r\cdot K_r\cdot t/(16\ln 2)} := n^{-\alpha r t}$$

where $\alpha = 25\cdot 3^q\cdot K_r/(16\ln 2)$. Finally, we count the number of choices. By Proposition 2, $|V(\Gamma)| \leq 150\cdot 3^q r\, n$. Thus each B_i has at most $\binom{|V(\Gamma)|}{r-1}+1 \leq (150\cdot 3^q r)^r n^r$ possible choices.

Therefore the number of well-behaved multisets $\mathcal{B}$ of size t is at most $(150\cdot 3^q r)^{rt}n^{rt}$. By Proposition 1, there are at most $N \leq \log\log n$ choices for k, $\log n$ choices for j.

Hence the probability that Lemma 2 fails for some fixed t is at most

$$(\log\log n)(\log n)\,(150\cdot 3^q r)^{rt}n^{rt}n^{-\alpha rt}.$$

By our choice of K_r, we have

$$r(1-\alpha) \leq -5,$$

then

$$(\log\log n)(\log n)\,(150\cdot 3^q r)^{rt}n^{rt}n^{-\alpha rt} \leq (150\cdot 3^q r)^{rt}(\log\log n)(\log n)\,n^{-5t} \leq (\log\log n)(\log n)n^{-4t}$$

using the assumption that $(150\cdot 3^q r)^r \leq n$, as n is sufficiently large.

Finally, taking a union bound over all $t \leq n$,

$$\mathbb{P}(\text{Lemma 2 fails}) \leq \sum_{t=1}^{n}(\log\log n)(\log n)n^{-4t} \leq (\log\log n)(\log n)n^{-4}\cdot n \leq n^{-1} = o(1).$$

This completes the proof. □

Theorem 2. *For $r \geq 3$, there exists a constant C_r such that for all sufficiently large n, there exists an r-uniform hypergraph Γ with*

$$e(\Gamma) \leq C_r n^{r-1}(\log n)^2(\log\log n)^r$$

which is universal for the class $\mathcal{H}^{(r)}(n,1)$ of n-vertex 1-degenerate r-graphs. In particular, one may take

$$C_r = 2(100\cdot 3^{r-1}r)^r.$$

Proof. Let $r \geq 3$. By Proposition 2, the random r-graph Γ defined in Definition 1 asymptotically almost surely satisfies

$$e(\Gamma) \leq C_r n^{r-1}(\log n)^2(\log\log n)^r,$$

and the statement of Lemma 2. It therefore remains to show that Γ is $\mathcal{H}^{(r)}(n,1)$-universal.

Let $H \in \mathcal{H}^{(r)}(n,1)$. Since H is 1-degenerate, there exists an ordering $v_1,\dots,v_n$ of $V(H)$ such that every vertex v_t has at most one back-link. We embed H into Γ using the embedding strategy from Definition 2.

Suppose, for a contradiction, that Γ is not $\mathcal{H}^{(r)}(n,1)$-universal. Then the embedding strategy must fail at some time t^* for some H. This means that there is no available candidate for v_{t^*}.

Let k^* be the positive integer chosen by the embedding strategy: we have $\Delta_{k^*} \leq d_H(v_{t^*}) \leq \Delta_{k^*-1}$ if $k^* < N$ and $d_H(v_{t^*}) \leq \Delta_{N-1}$ if $k^* = N$, where $d_H(v_{t^*})$ is the degree of v_{t^*} in H.

Let $B := \psi_{t^*-1}(L^-(v_{t^*}))$, notice that B is empty or contains a single $(r-1)$-set.

Claim 1. *$\{B\}$ is a well-behaved multiset.*

Proof. If $B = \emptyset$, then $\{B\}$ is a well-behaved multiset straightforwardly. If $|B| = 1$, condition *(WB1)* is immediate. For *(WB2)*, let $1 \le k \le N$ and let $u \in W_k$. Since $|B| = 1$, $\big|\{\, i \in [1] : u \in V(B_i)\,\}\big| \le 1 < 3 \le \Delta_N \le \Delta_{k-1}$, thus *(WB2)* holds.

For *(WB3)*, let $1 \le k \le N$ and $1 \le j \le \log n$. Since B consists of a single $(r-1)$-set, $|V(B) \cap W_{k,j}| \le r-1$. On the other hand,

$$|W_{k,j}| \ge \frac{|W_k|}{2\log n} = \frac{50 \cdot 3^q r n}{\Delta_k \log n} = \frac{50 \cdot 3^q r\, n^{1-q^{-k}}}{\log n}.$$

Since $q = r - 1 \ge 2$, we have $1 - q^{-k} > 0$. For all sufficiently large n,

$$|V(B) \cap W_{k,j}| \le r - 1 \le \frac{1}{2}\frac{50 \cdot 3^q r\, n^{1-q^{-k}}}{\log n} \le \frac{1}{2}|W_{k,j}|.$$

Hence (WB3) also holds. Therefore $\{B\}$ is well-behaved. □

Applying Lemma 2 with the well-behaved multiset $\{B\}$, $k = k^*$, and $j = \log n$, we obtain

$$\begin{aligned}
\big|\{u \in W_{k^*,\log n} : B \subseteq L(u)\}\big| &\ge \min\left\{\frac{1}{16}, \frac{1}{8}p^*\Delta_{k^*}\right\}|W_{k^*,\log n}| \\
&\ge \min\left\{\frac{1}{16} \cdot \frac{100 \cdot 3^q rn}{2\Delta_{k^*}\log n}, \frac{1}{8} \cdot \frac{100 \cdot 3^q rn}{2\Delta_{k^*}\log n} \cdot \frac{K_r(\log n)^2}{n}\Delta_{k^*}\right\} \\
&= \min\left\{\frac{100 \cdot 3^q r}{32} \cdot \frac{n^{1-q^{-k^*}}}{\log n}, \frac{50 \cdot 3^q r K_r}{8}\log n\right\} \\
&> 1
\end{aligned}$$

The last inequality uses the facts that $1 - q^{-k^*} > 0$ and both quantities tend to infinity for all sufficiently large n. Thus there exists more than one candidate for v_{t^*} in $W_{k^*,\log n}$. Since the embedding strategy fails at time t^*, every such candidate must belong to $\mathrm{Im}(\psi_{t^*-1})$.

Define

$$L_{k,j} := (4\log n)^{1-j}\begin{cases}\dfrac{rn}{\Delta_k}, & 1 \le k \le N-1, \\ n, & k = N.\end{cases}$$

We make the following assertion: for every $t \le t^* - 1$, $|\,\mathrm{Im}(\psi_t) \cap W_{k,j}| \le L_{k,j}$ for all $1 \le k \le N$ and $1 \le j \le \log n$.

Assuming this is true, for $k^* < N$, we obtain

$$\begin{aligned}
|\,\mathrm{Im}(\psi_{t^*-1}) \cap W_{k^*,\log n}| &\le L_{k^*,\log n} \\
&= (4\log n)^{1-\log n}\frac{rn}{\Delta_{k^*}} \\
&= (4\log n)^{1-\log n} rn^{\,1-q^{-k^*}} \\
&= 4r\log n \cdot n^{\,1-q^{-k^*}-\log(4\log n)} \\
&= 4r\log n \cdot n^{\,1-q^{-k^*}-2-\log\log n} \\
&\le 4r\log n \cdot n^{-1} \\
&\le 1,
\end{aligned}$$

and for $k^* = N$, we obtain

$$\begin{aligned}
|\,\mathrm{Im}(\psi_{t^*-1}) \cap W_{k^*,\log n}| &\le L_{k^*,\log n} \\
&= (4\log n)^{1-\log n} n \\
&\le 1,
\end{aligned}$$

for both cases, we used the assumption that n is sufficiently large at the end. If the claim holds, then not all candidates in $W_{k^*,\log n}$ can be covered by $\mathrm{Im}(\psi_{t^*-1})$, contradicting the assumption that the embedding strategy fails at time t^*.

It therefore remains to prove the assertion. We proceed by induction on t.

For the base case $t = 1$, the embedding strategy embeds v_1 into some vertex of $W_{k,1}$, since $L^-(v_1) = \emptyset$. Hence $|\mathrm{Im}(\psi_1) \cap W_{k,1}| = 1 \le L_{k,1}$, and $|\mathrm{Im}(\psi_1) \cap W_{k,j}| = 0 < L_{k,j}$ for every $j > 1$. Thus the claim holds for $t = 1$.

Now let $1 < t \le t^* - 1$, and assume that $|\mathrm{Im}(\psi_{t-1}) \cap W_{k,j}| \le L_{k,j}$ for every k and j. We will show that $|\mathrm{Im}(\psi_t) \cap W_{k,j}| \le L_{k,j}$ for every k and j.

Suppose not. Since the only difference between $\mathrm{Im}(\psi_t)$ and $\mathrm{Im}(\psi_{t-1})$ is the newly embedded vertex $\psi_t(v_t)$, there exists a unique pair (k', j') such that $\psi_t(v_t) \in W_{k',j'}$ and

$$|\mathrm{Im}(\psi_t) \cap W_{k',j'}| > L_{k',j'}. \tag{A1}$$

For every other pair $(k, j) \neq (k', j')$, we still have

$$|\mathrm{Im}(\psi_t) \cap W_{k,j}| \le L_{k,j}. \tag{A2}$$

We will show that this is impossible.

Case 1: $j' = 1$**.**

If $k' = N$, then

$$|\mathrm{Im}(\psi_t) \cap W_{N,1}| > L_{N,1} = n,$$

which is impossible since H has only n vertices and $t \le n$.

Now suppose $k' < N$. Recall that a vertex of H is embedded into $W_{k'}$ only if its degree is greater than $\Delta_{k'}$. By Observation 1, the number of such vertices is at most $\frac{rn}{\Delta_{k'}}$. Therefore

$$|\mathrm{Im}(\psi_t) \cap W_{k',1}| \le \frac{rn}{\Delta_{k'}} = L_{k',1},$$

contradicting assumption A1.

Case 2: $j' > 1$**.** Let

$$X_{k',j'} := \{x \in \{v_1, \dots, v_t\} : \psi_t(x) \in W_{k',j'}\},$$

and define the multiset

$$\mathcal{B} := (B_x)_{x \in X_{k',j'}}, \text{where } B_x := \psi_t(L^-(x)).$$

Then

$$|\mathcal{B}| = |X_{k',j'}| = |\mathrm{Im}(\psi_t) \cap W_{k',j'}|.$$

Claim 2. *$\mathcal{B}$ is well-behaved.*

Proof. Condition (WB1) holds since every back-link has size at most 1, and ψ_t is an embedding. For (WB2), let $u \in W_k$. Suppose $u \in V(B_i)$ for some i. Then there exists a vertex $v_b \in V(H)$ such that $u \in \psi_t(L^-(v_b))$. Since $u \in \mathrm{Im}(\psi_t)$, there exists a vertex v_a with $\psi_t(v_a) = u$. As u belongs to the embedded back-link of v_b, we have $a < b$. Therefore

$$\big|\{i : u \in V(B_i)\}\big| \le d_H(v_a) \le \Delta_{k-1}$$

by the definition of embedding strategy. Hence (WB2) holds.

For (WB3), by the induction hypothesis we have $|\mathrm{Im}(\psi_{t-1}) \cap W_{k,j}| \le L_{k,j} + 1$ for every k, j. We shall show that $L_{k,j} + 1 \le \frac{1}{16}|W_{k,j}|$ for every k, j.

First consider $j = 1$. If $k = N$, then $|W_{N,1}| = \frac{1}{2}|W_N| \ge 50rn$. Since $L_{N,1} = n$, $L_{N,1} + 1 = n + 1 < \frac{50}{16}rn \le \frac{1}{16}|W_{N,1}|$ for sufficiently large n. Now suppose $k < N$. Then $|W_{k,1}| = \frac{1}{2}|W_k| = \frac{50 \cdot 3^q rn}{\Delta_k}$, and $L_{k,1} + 1 = \frac{rn}{\Delta_k} + 1 < \frac{50 \cdot 3^q rn}{16\Delta_k} = \frac{1}{16}|W_{k,1}|$ for sufficiently large n.

Now consider $j > 1$. Since $L_{k,j} \le (4 \log n)^{1-j} L_{k,1} \le \frac{L_{k,1}}{4 \log n}$, and $|W_{k,j}| \ge \frac{|W_{k,1}|}{\log n}$, we obtain

$$L_{k,j} + 1 \le \frac{L_{k,1}}{4 \log n} + 1 \le \frac{|W_{k,1}|}{64 \log n} + 1 \le \frac{|W_{k,j}|}{64} + 1$$

Furthermore, $|W_{k,j}| \geq \frac{50 \cdot 3^q rn}{\Delta_k \log n} = \frac{50 \cdot 3^q r\, n^{1-q^{-k}}}{\log n}$, which tends to infinity as $n \to \infty$. Hence, for sufficiently large n, $\frac{|W_{k,j}|}{64} + 1 \leq \frac{1}{16}|W_{k,j}|$. Therefore $L_{k,j} + 1 \leq \frac{1}{16}|W_{k,j}|$, and (WB3) follows. □

We may therefore apply Lemma 2 to $\mathcal{B}$ with $k = k'$ and $j = j' - 1$, which gives

$$\begin{aligned}
&\Big|\{u \in W_{k',j'-1} : \exists B \in \mathcal{B},\ B \subseteq L(u)\}\Big| \\
&\geq \min\left\{\frac{1}{16}, \frac{1}{8}|\mathcal{B}|p^*\Delta_{k'}\right\}|W_{k',j'-1}| \\
&\geq \min\left\{\frac{|W_{k',j'-1}|}{16}, \frac{1}{8}|\operatorname{Im}(\psi_t) \cap W_{k',j'}|p^*\Delta_{k'}|W_{k',j'-1}|\right\} \\
&\geq \min\Big\{L_{k',j'-1} + 1, \frac{1}{8}L_{k',j'}p^*\Delta_{k'}|W_{k',j'-1}|\Big\} \\
&\geq \min\Big\{L_{k',j'-1} + 1, \frac{1}{8}\frac{L_{k',j'-1}}{4\log n} \cdot \frac{K_r(\log n)^2}{n} \cdot \Delta_{k'} \cdot \frac{100 \cdot 3^q rn}{2\Delta_{k'}\log n}\Big\} \\
&\geq \min\Big\{L_{k',j'-1} + 1, \frac{100 \cdot 3^q rK_r}{64}L_{k',j'-1}\Big\}
\end{aligned}$$

By the choice of K_r, we have

$$\frac{100 \cdot 3^q rK_r}{64} > 1.$$

Therefore both terms in the minimum are strictly greater than $L_{k',j'-1} \geq |\operatorname{Im}(\psi_{t-1}) \cap W_{k',j'-1}|$.

Hence there exists

$$u \in \{u \in W_{k',j'-1} : \exists B \in \mathcal{B},\ B \subseteq L(u)\} \setminus \operatorname{Im}(\psi_{t-1}).$$

By the definition, there is a vertex $x = v_s$ such that $B_x \in \mathcal{B}$ and $B_x \subseteq L(u)$. Since every vertex occurring in $L^-(x)$ precedes $x = v_s$, we have

$$B_x = \psi_t(L^-(x)) = \psi_{s-1}(L^-(x)).$$

Moreover, $u \notin \operatorname{Im}(\psi_{t-1})$ implies $u \notin \operatorname{Im}(\psi_{s-1})$. Thus u was an available candidate for x in $W_{k',j'-1}$ when x was embedded. This contradicts the minimality of j', since x was embedded into $W_{k',j'}$.

□

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